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International Mathematics Assessments for Schools (2025-2026)

Junior Division Round 2

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the **answer sheet**. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the **answer sheet** to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

Junior Division Round 2

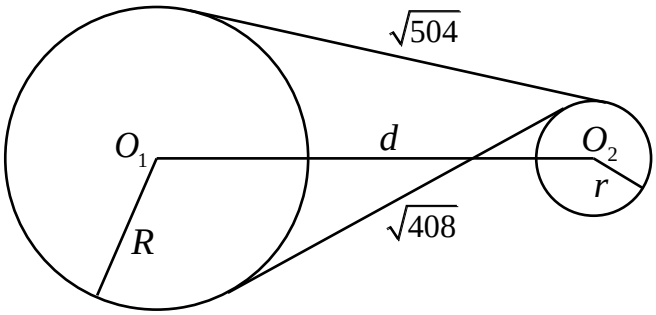
Questions 1 to 5, 4 marks each

1. Let *a*, *b* and *c* be positive integers such that $a^2+b^2=c^2$, where $0<a<b<c$. The ordered triple (*a*, *b*, *c*) is called a Pythagorean triple. For example, (3, 4, 5) is a Pythagorean triple. How many Pythagorean triples (*a*, *b*, *c*) are there such that $a+b+c=120$?

（A）1　　（B）2　　（C）3　　（D）4　　（E）5

---

2. Let O_1 and O_2 be two disjoint circles with integer radii *R* and *r*, in cm, respectively, where $R>r$. The distance *d*, in cm, is the distance between the centres of O_1 and O_2 is also an integer. It is known that the length of each external common tangent of O_1 and O_2 is $\sqrt{504}$ cm, and the length of each internal common tangents of O_1 and O_2 is $\sqrt{408}$ cm. What is the value of *d*?



（A）22　　（B）23　　（C）24　　（D）25　　（E）26

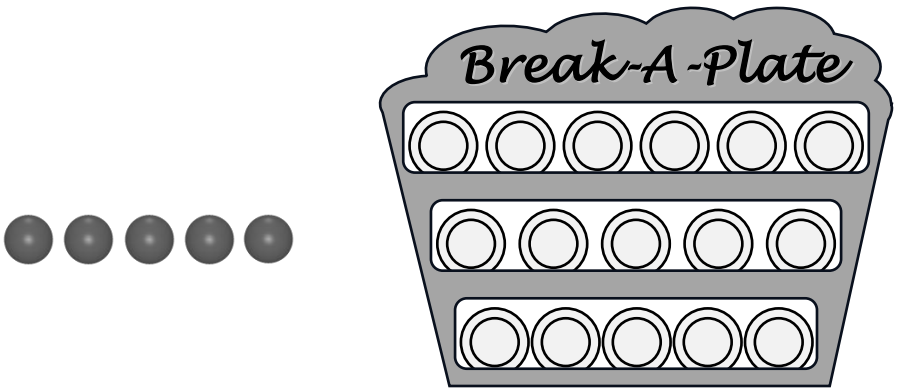
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3. Consider the sequence of positive integers that are relatively prime to 210, arranged in increasing order. What is the 2026th term of this sequence?

（A）8853　　（B）8857　　（C）8859　　（D）8861　　（E）8863

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4. In the carnival game *Break-A-Plate*, a player throws five softballs at plates from a distance. If the player successfully breaks plates on any two consecutive throws among the five throws, the player wins a prize.



Suppose that for each throw, Charlie has a $\frac{2}{3}$ chance of breaking a plate, and that all throws are independent of each other, what is the probability that Charlie wins a prize?

(A) $\frac{128}{243}$ (B) $\frac{190}{243}$ (C) $\frac{192}{243}$ (D) $\frac{200}{243}$ (E) $\frac{225}{243}$

---

5. Students are divided into many teams to participate in a contest in two different halls. If 120 students have exactly two teammates in the other hall, 160 students have at least one other teammate in each of the two halls and no three teammates are in the same hall, how many students have exactly one other teammate in each hall?

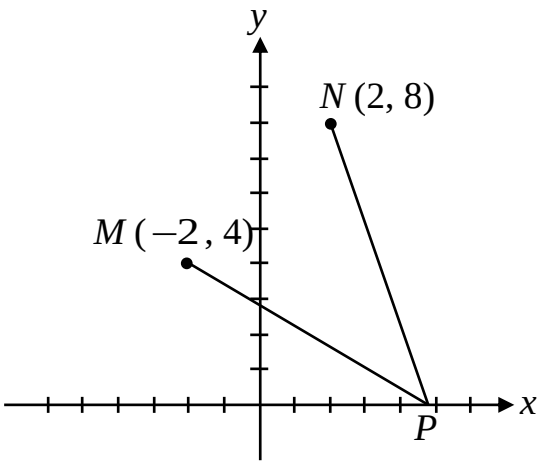
(A) 32 (B) 40 (C) 60 (D) 64 (E) 80

Questions 6 to 13, 5 marks each

6. Remove the trailing zeros from 2025! and 2026! and let the resulting numbers be A and B, respectively. Let a and b be the remainders when A and B are divided by 100, respectively. What is the value of $|a-b|$?

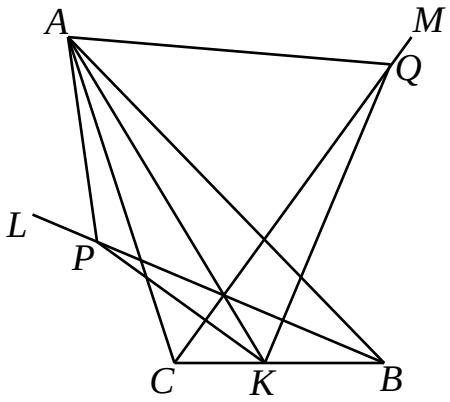
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7. Let P be a point on the x-axis such that $\angle MPN$ attains its maximum possible value, where point M is at $(-2, 4)$ and point N is at $(2, 8)$. What is the x-coordinate of P?



8. Let a and b be positive integers such that $\frac{1}{a}+\frac{1}{b}>\frac{1}{5}$. If $a \geq 6$ and $b \geq 6$, what is the minimum possible value of $\frac{1}{a}+\frac{1}{b}$?

9. As shown in the diagram below, AK, BL and CM are the angle bisectors of triangle ABC, with K lying on side BC. Points P and Q are on lines BL and CM, respectively, such that $AP = PK$ and $AQ = QK$. If $\angle BAC = 26°$, what is the angle measure, in degrees, of $\angle PAQ$?



10. In three-dimensional space, *ABC* is an equilateral triangle with side length 1 cm. Let *D* be a point in space such that $AD = 10$ cm and the lengths, in cm, of both *BD* and *CD* are integers. What is the sum of the lengths, in cm, of *BD* and *CD*?

11. What is the smallest positive integer *y* such that $2026y^2 + 1$ is a perfect square?

12. It is known that every positive integer *n* can be uniquely written as a sum of one or more non-consecutive Fibonacci numbers, $F_2 = 1$, $F_3 = 2$, $F_4 = 3$, $F_5 = 5$, …, excluding F_0 and F_1.

For example, we can represent 30 as $1 + 8 + 21 = F_2 + F_6 + F_8$.

Let a_n be the number of terms in the Fibonacci representation of *n*, for example, $a_{30} = 3$. What is the sum $a_1 + a_2 + ... + a_{54}$?

13. How many non-empty subsets of the set {1, 2, 3, …, 11} contain no two consecutive numbers?

ANSWER SHEET of Questions 1 to 13

Questions 1 to 5, 4 marks each

- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct.

1.		**2.**		**3.**		**4.**		**5.**	

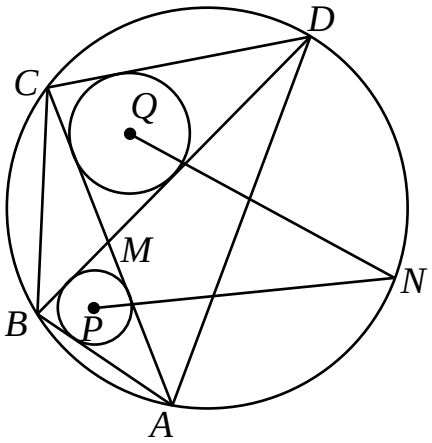
Questions 6 to 13, 5 marks each

- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited.

6.		**10.**	cm
7.		**11.**	
8.		**12.**	
9.	○	**13.**	subsets

Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

14. As shown in the diagram below, *ABCD* is a cyclic quadrilateral, where diagonals *AC* and *BD* intersect at *M*. Let *N* be the midpoint of the arc *AD*. If *P* and *Q* are the incentres of triangles *ABM* and *CDM*, respectively, and $PN = 3\sqrt{3}$ cm, what is the length, in cm, of *QN*?



Answer : _____ cm

15. Five cards, numbered 1 to 5, are arranged randomly in a row. They are sorted from smallest to largest using the "Bubble Sort" algorithm, which is defined as follows: starting from the left, compare the first two adjacent cards and swap them if the left card is bigger; then move one position to the right and repeat this process until reaching the end of the row. This procedure is repeated from the beginning until the cards are fully sorted.

Count every swap made during the entire process. For example, the arrangement (4, 5, 3, 2, 1) requires 9 swaps to be fully sorted, as shown below:

(4, **5**, **3**, 2, 1) → (4, 3, **5**, **2**, 1) → (4, 3, 2, **5**, **1**) → (**4**, **3**, 2, 1, 5)
→ (3, **4**, **2**, 1, 5) → (3, 2, **4**, **1**, 5) → (**3**, **2**, 1, 4, 5) → (2, **3**, **1**, 4, 5)
→ (**2**, **1**, 3, 4, 5) → (1, 2, 3, 4, 5)

How many initial arrangements require exactly 5 swaps to be fully sorted?

Answer : _____ initial arrangements